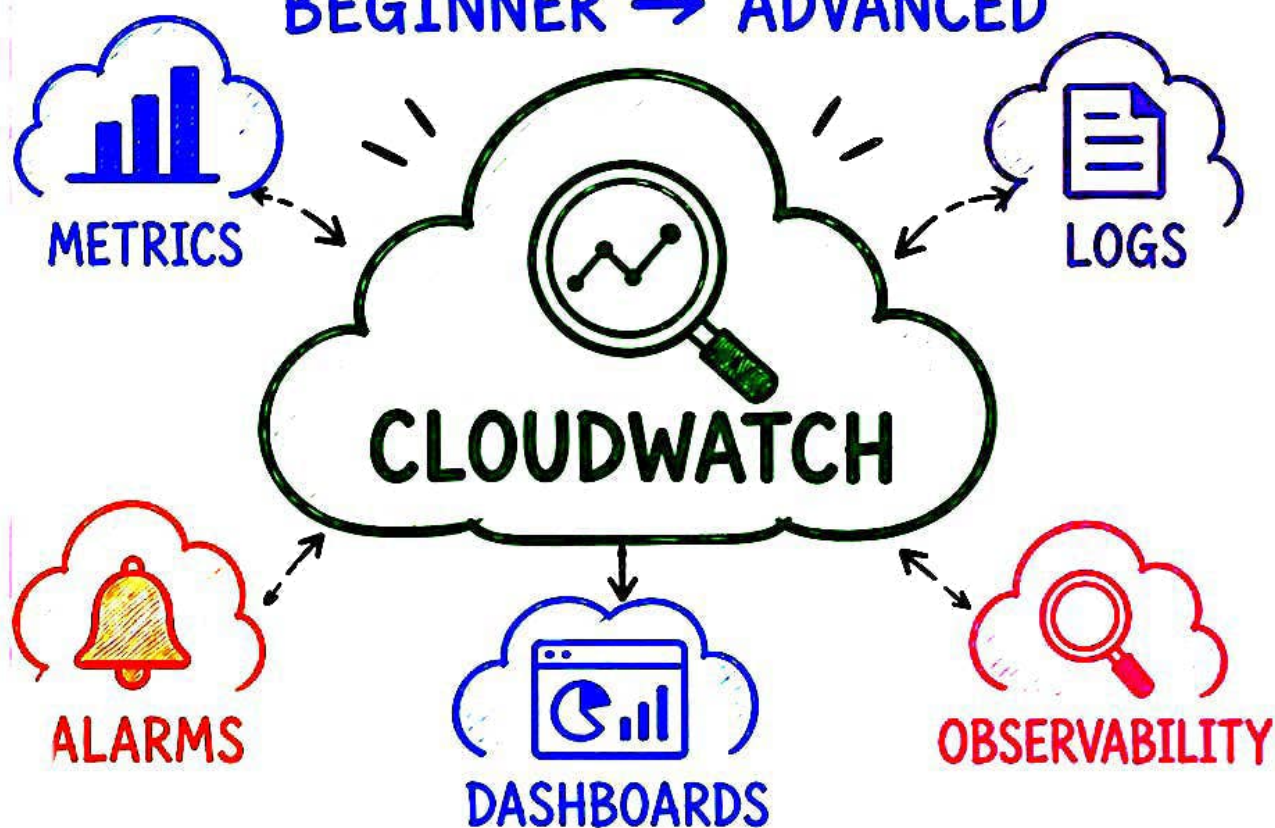


AWS CLOUDWATCH

COMPLETE NOTES

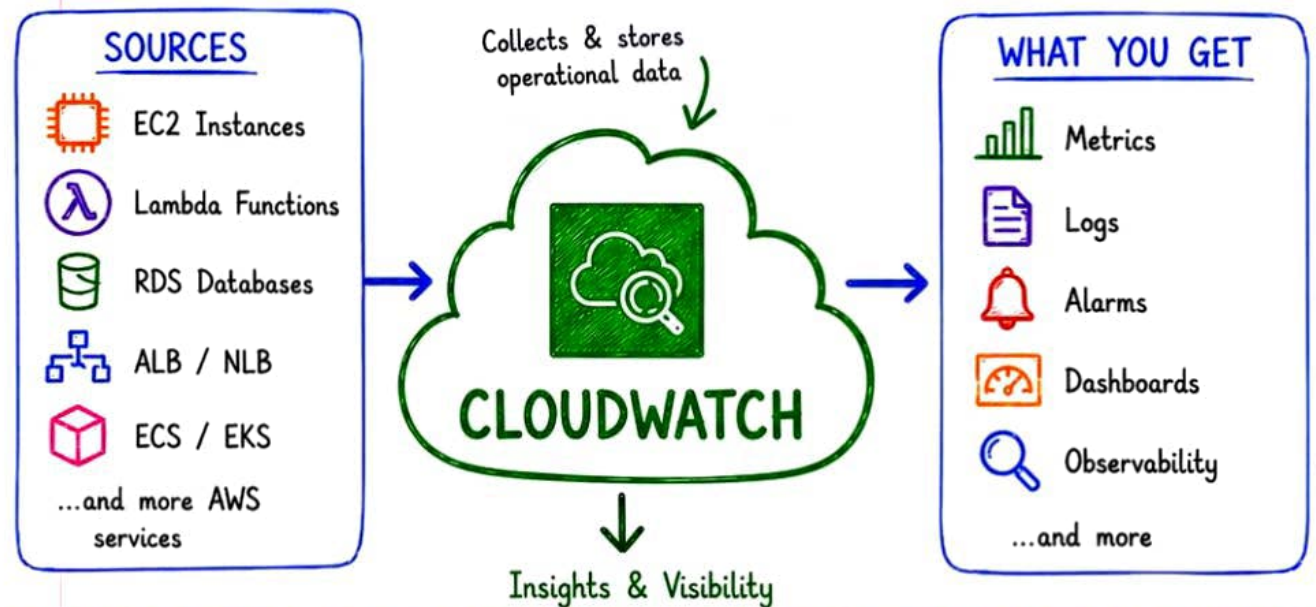
BEGINNER → ADVANCED



EVERYTHING YOU NEED TO KNOW
ABOUT AWS CLOUDWATCH

WHAT IS AMAZON CLOUDWATCH?

Amazon CloudWatch is AWS's **monitoring** and **observability** service for applications and infrastructure.



MEMORY TRICK

CloudWatch helps you
OBSERVE → DETECT →
INVESTIGATE → ACT ☆

CloudWatch =
Eyes & Ears
of Your AWS Environment



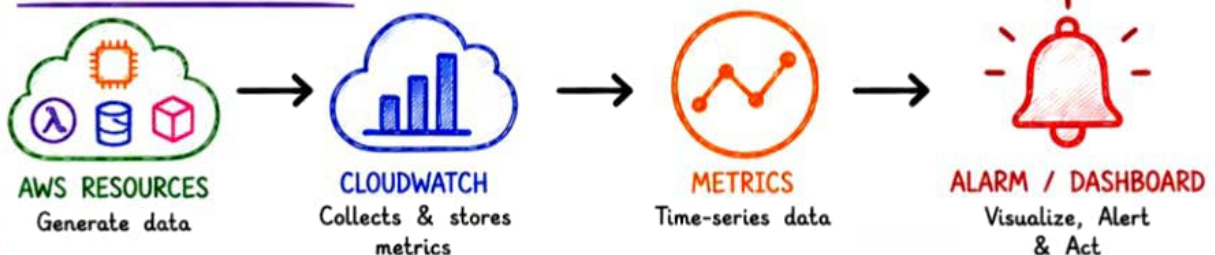
From infrastructure metrics to application logs,
CloudWatch gives you **complete visibility** into your AWS workloads.

③ = CLOUDWATCH METRICS =

A Metric is a numerical measurement of a resource collected over time.



HOW METRICS WORK?



EXAMPLES OF CLOUDWATCH METRICS

EC2 METRICS



- CPUUtilization
- NetworkIn / NetworkOut
- DiskReadOps / DiskWriteOps
- DiskSpaceUtilization
- StatusCheckFailed
- ...and more

MemoryUtilization is **NOT** a default metric.
Use CloudWatch Agent.

LAMBDA METRICS



- Invocations
- Errors
- Duration
- Throttles
- ConcurrentExecutions
- ...and more

Helps monitor performance & troubleshoot serverless applications.

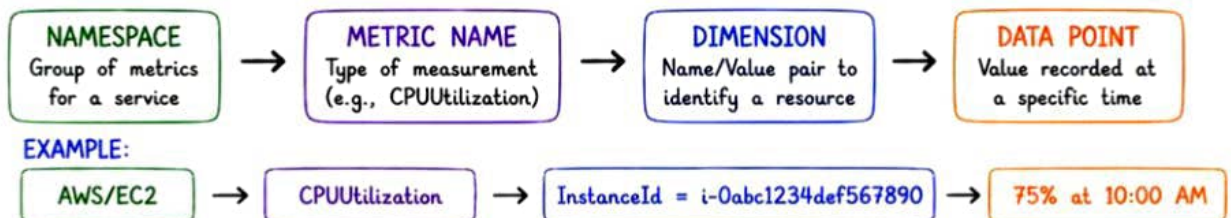
ALB / NLB METRICS



- RequestCount
- TargetResponseTime
- HTTPCode_Target_5XX_Count
- HealthyHostCount
- UnHealthyHostCount
- ...and more

Use these metrics to monitor traffic, latency & errors.

★ NAMESPACES, METRICS & DIMENSIONS



KEY TAKEAWAY

- ✓ Metrics provide real-time visibility.
- ✓ Every metric is a time-series data.
- ✓ Different services have different metrics.
- ✓ Use the right metrics to monitor performance, availability & errors.



IMPORTANT NOTE



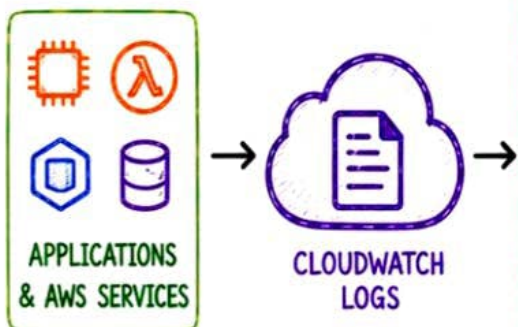
EC2 memory metrics are not available by default. Install CloudWatch Agent or send custom metrics.



4 CLOUDWATCH LOGS

CloudWatch Logs lets you **collect**, **store** and **analyze** log data from your applications and AWS resources.

LOG DATA FLOW



LOG STRUCTURE



EXAMPLE

Log Group:
/aws/lambda/payment-function
↓
Log Stream:
2026/08/29/[\$LATEST]abcd123
↓
Log Event:
2026-08-29 10:15:21
ERROR: Payment processing failed for order 12345

CLOUDWATCH LOGS INSIGHTS

Query and analyze large amounts of log data in real-time.

- ☆ Search by keywords
- ☆ Filter by fields
- ☆ Aggregate & visualize
- ☆ Faster troubleshooting



COMMON USE CASES

- ✓ Troubleshoot application errors
- ✓ Audit & compliance logs
- ✓ Security & access logs
- ✓ API & request tracking
- ✓ Operational & debug logs
- ✓ Centralized log management



KEY TAKEAWAY

Metrics tell **WHAT** happened.
Logs help explain **WHY** it happened.

BEST PRACTICES



USE MEANINGFUL LOG GROUP NAMES
e.g., /aws/lambda/my-function



SET LOG RETENTION
Delete old logs & control costs.



SECURE LOG ACCESS
Use IAM policies & encryption.



USE LOGS INSIGHTS
Speed up search & troubleshooting.



MONITOR LOG COSTS
Track ingestion, storage & query costs.



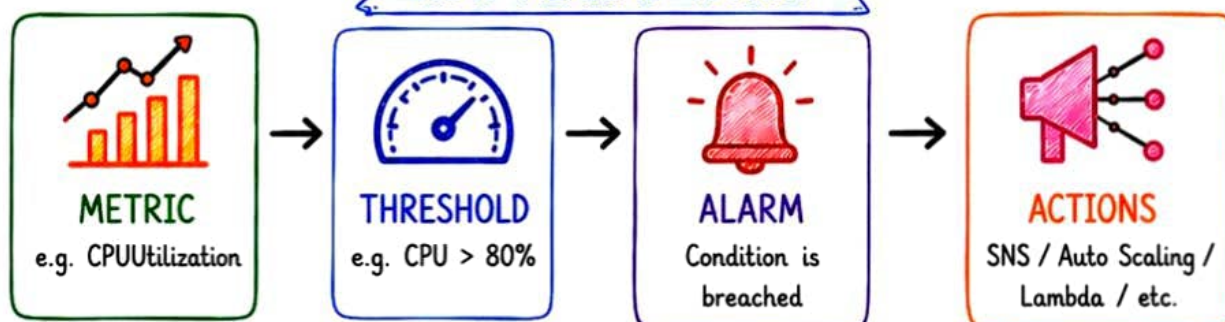
CENTRALIZE • ANALYZE • UNDERSTAND • ACT



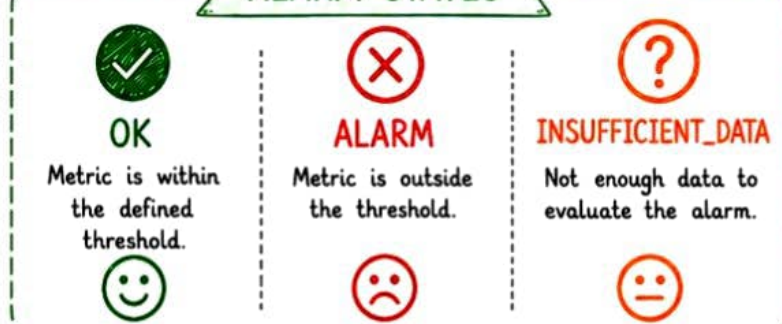
5 CLOUDWATCH ALARMS

CloudWatch Alarms **monitor** your metrics and **notify** you or **take action** when conditions are met.

HOW ALARMS WORK?



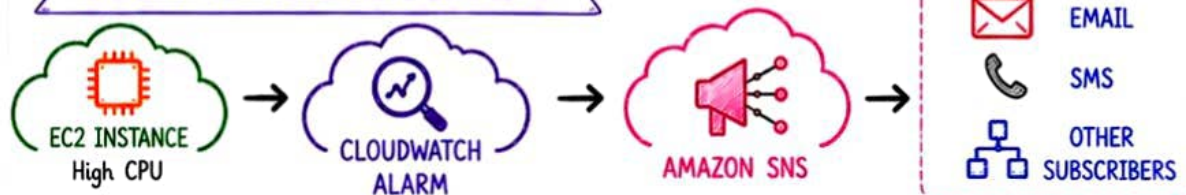
ALARM STATES



EXAMPLE

Metric: CPUUtilization (EC2 Instance)
Condition: CPU > 80% for 5 minutes
Evaluation Periods: 5
Datapoints to Alarm: 3 out of 5
Action: Notify via SNS

ALARM WITH SNS NOTIFICATIONS



COMMON ALARM ACTIONS

- ✓ Send notification via SNS (Email / SMS)
- ✓ Trigger Auto Scaling actions
- ✓ Invoke AWS Lambda function
- ✓ Stop / Start AWS EC2 instances
- ✓ Create incident in ITSM tools

BEST PRACTICES

- ✓ Set meaningful thresholds.
- ✓ Avoid too many alarms (alarm fatigue).
- ✓ Use appropriate evaluation periods.
- ✓ Test your alarms regularly.
- ✓ Use descriptive alarm names.
- ✓ Monitor both infrastructure & application metrics.

★ MONITOR → DETECT → ALERT → ACT ★



CloudWatch Alarms help you detect issues early and keep your applications healthy!



6 CLOUDWATCH DASHBOARDS

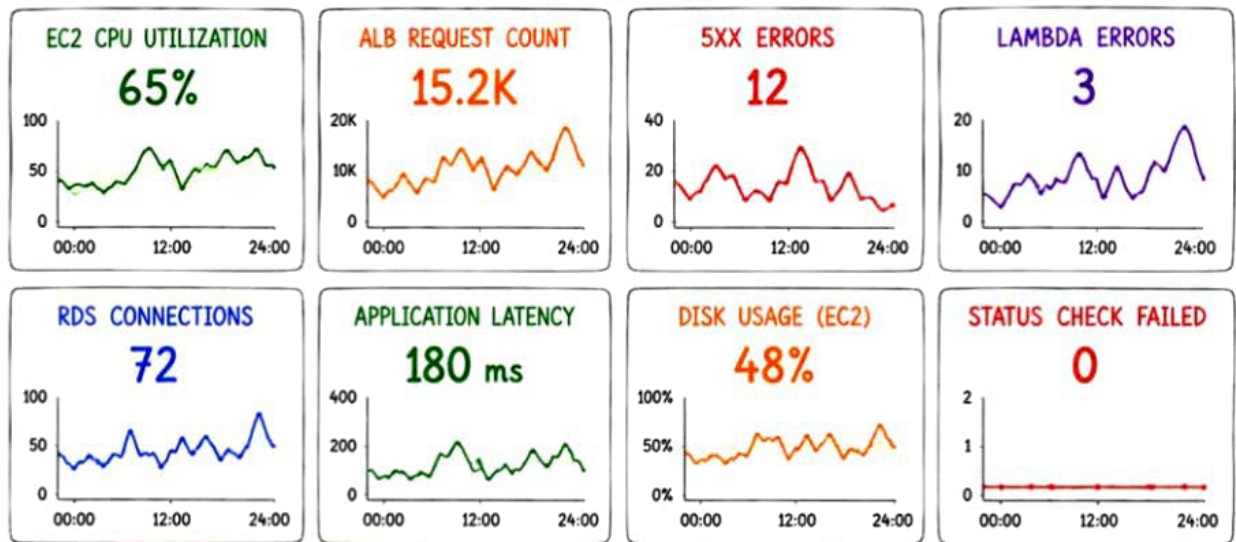
Dashboards give you a **single, real-time view** of your AWS resources and application health.



WHAT IS A DASHBOARD?

A dashboard is a **customizable home** for your CloudWatch metrics, logs insights, alarms, and other AWS data.

PRODUCTION DASHBOARD (EXAMPLE)



BENEFITS

- ✓ Centralized view of your environment
- ✓ Real-time monitoring of important metrics
- ✓ Faster troubleshooting & root cause analysis
- ✓ Track performance, availability & errors
- ✓ Share with your team or stakeholders



CUSTOMIZATION OPTIONS

- Add widgets (metrics, logs, alarms)
- Filter by time range, region, resources
- View alarm status directly
- Pin important dashboards for quick access



MEMORY TRICK:

WHAT

+

WHERE

+

WHEN

=

DASHBOARD

Know **What** is happening, **Where** it is happening, and **When** it happened.



ONE DASHBOARD = COMPLETE VISIBILITY



7 CLOUDWATCH AGENT & CUSTOM METRICS

Standard AWS metrics do not include everything.
CloudWatch Agent & Custom Metrics fill the gap!

CLOUDWATCH AGENT

The CloudWatch Agent collects **additional** system-level metrics and logs from your EC2 instances, on-prem servers, and containers.



WHAT IT CAN COLLECT

- ✓ Memory Utilization
- ✓ Disk Usage / IO
- ✓ CPU / Processes
- ✓ Network Metrics
- ✓ Log Files
- ✓ Custom Application Logs

Install agent on your instance and configure it to send data to CloudWatch.

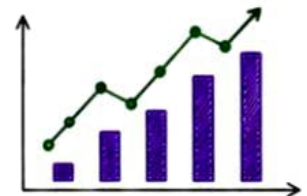
CUSTOM METRICS

Publish your own application/business metrics to CloudWatch using AWS SDK, CLI, or CloudWatch API.



EXAMPLES

- OrdersProcessed
- PaymentFailures
- ActiveUsers
- QueueDepth
- CartAbandonments



WHY THEY ARE IMPORTANT?



Get deeper visibility beyond default AWS metrics



Detect issues early with detailed system metrics



Optimize performance and reduce operational cost



Monitor application health with business metrics that matter



Better troubleshooting with logs + metrics in one place

INTERVIEW HIGHLIGHT

EC2 memory utilization is **NOT** a standard metric. You need **CloudWatch Agent** or a custom metric to monitor it.



BEST PRACTICES

- ✓ Install and configure CloudWatch Agent properly.
- ✓ Collect only required metrics to control costs.
- ✓ Use meaningful custom metrics for your business.
- ✓ Set alarms on important custom metrics.
- ✓ Centralize logs and metrics for better observability.

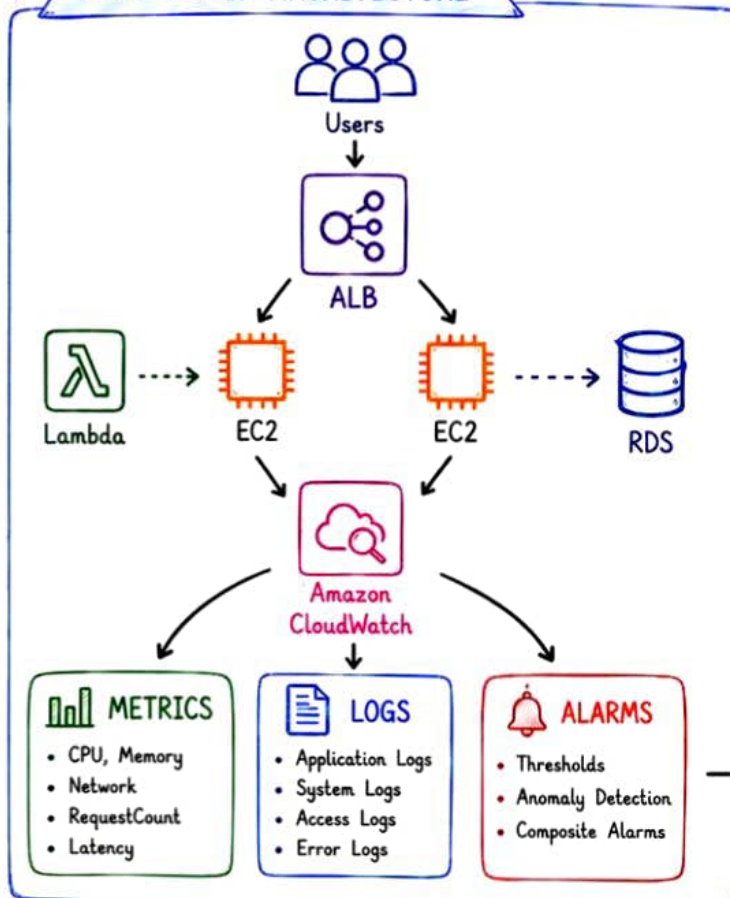


MORE DATA → BETTER INSIGHTS → SMARTER DECISIONS

8 CLOUDWATCH IN PRODUCTION

CloudWatch integrates with all your AWS resources to give end-to-end monitoring, logging and alerting.

REAL-WORLD ARCHITECTURE



What CloudWatch Monitors?

- ✓ Infrastructure (EC2, RDS, VPC, ALB...)
- ✓ Applications (Lambda, ECS, EKS...)
- ✓ Logs & Events
- ✓ Custom Business Metrics
- ✓ User Activity & Performance

KEY BENEFIT

One central place to monitor, detect and respond to issues in real-time.

CLOUDWATCH INTEGRATIONS



SNS

Send alerts via Email / SMS / Push / ChatOps



LAMBDA

Trigger automation & remediation workflows



AUTO SCALING

Scale resources based on CloudWatch metrics & alarms



DASHBOARDS

Visualize metrics & system health in one view



IAM

Secure access to metrics, logs & dashboards



TAKEAWAY → CloudWatch is the central nervous system of your AWS environment.

≡ MONITOR • DETECT • ALERT • AUTOMATE ≡



9 CLOUDWATCH PRICING & COST TIPS

CloudWatch is powerful, but costs can grow fast.
Understand pricing and follow best practices to control **costs**.

CLOUDWATCH PRICING MODEL



METRICS

\$0.30 per metric per month
(Standard Resolution)



CUSTOM METRICS

\$0.30 per custom metric per month



LOGS INGESTION

\$0.50 per GB ingested



LOGS ARCHIVAL STORAGE

\$0.03 per GB per month



ALARMS

\$0.10 per alarm per month
(First 10 alarms free)



DASHBOARDS

\$3.00 per dashboard per month
(First 3 dashboards free)



DATA TRANSFER

Charges may apply
for cross-region / cross-account

GOOD TO KNOW

- First 10 custom metrics free
- First 5GB logs ingestion free (monthly)
- First 10 alarms free
- First 3 dashboards free



WHERE COSTS COME FROM?

- ✓ High number of custom metrics



- ✓ Verbose logs (too much data)



- ✓ High log retention period



- ✓ Many alarms & dashboards



- ✓ Cross-region / cross-account usage



COST OPTIMIZATION TIPS

- ✓ Use Standard metrics instead of Custom metrics



- ✓ Filter & reduce unnecessary logs



- ✓ Set appropriate log retention



- ✓ Delete unused dashboards & alarms



- ✓ Use metric math & anomaly detection to reduce alarms



- ✓ Archive old logs to S3 / Glacier



- ✓ Monitor CloudWatch usage & costs regularly



COST SAVING STRATEGY ROADMAP



IDENTIFY

Identify high cost metrics & log groups



ANALYZE

Analyze usage patterns & retention



OPTIMIZE

Optimize metrics, alarms, retention & filters



MONITOR

Monitor costs in Cost Explorer regularly



REVIEW

Review & iterate for continuous improvement



PAY ONLY FOR WHAT YOU USE.

Monitor **smartly**. Log **wisely**. Save **consistently**.

≡ **MONITOR** • **OPTIMIZE** • **SAVE** ≡



10 CLOUDWATCH BEST PRACTICES

Use CloudWatch the right way to get maximum visibility,
reduce noise, and improve response time.

1. PLAN WHAT TO MONITOR

- ✓ Identify critical applications & infrastructure.
- ✓ Define important metrics & logs.
- ✓ Set meaningful thresholds.
- ✓ Monitor business KPIs, not just system metrics.



2. USE STANDARD RESOLUTION

- High Resolution (1 sec):
Detailed monitoring, higher cost.
- Standard Resolution (60 sec):
Default & cost-effective for most use cases.



3. SET SMART ALARMS

- ✓ Use metric math to create better alarms.
- ✓ Avoid basic CPU/Memory alarms.
- ✓ Use anomaly detection for dynamic baselines.
- ✓ Suppress alarms during maintenance windows.
- ✓ Include clear alarm names & descriptions.



4. STRUCTURE DASHBOARDS

- ✓ Create role-based dashboards.
- ✓ Keep dashboards simple & focused.
- ✓ Use widgets that add real value.
- ✓ Avoid overcrowding dashboards.
- ✓ Share dashboards with your team.



5. CENTRALIZE LOGS

- ✓ Send all logs to CloudWatch Logs.
- ✓ Use consistent log groups & naming.
- ✓ Set retention policies to control costs.
- ✓ Use log filters & pattern searches.
- ✓ Archive old logs to S3 / Glacier.



6. CONTROL COSTS

- ✓ Use metric filters & aggregation wisely.
- ✓ Delete unused alarms & dashboards.
- ✓ Reduce log retention period.
- ✓ Avoid high-resolution metrics unless needed.
- ✓ Review usage in Cost Explorer regularly.



7. IMPROVE OBSERVABILITY



Correlate logs, metrics & alarms together.



Integrate with SNS / Slack / PagerDuty.



Use Lambda for auto remediation.



Track trends with CloudWatch Insights.



Review reports & insights regularly.



Collaborate with DevOps & Ops teams.

ULTIMATE CHEAT SHEET



- ✓ **PLAN** → Identify & Prioritize
- ✓ **MONITOR** → Metrics, Logs & Alarms
- ✓ **ANALYZE** → Insights & Trends
- ✓ **OPTIMIZE** → Improve & Reduce Costs

Golden Rule:
Monitor what matters,
alert smartly, and
optimize continuously.



GOOD MONITORING = FASTER RESPONSE = HAPPY USERS = LOWER COSTS

